# IND-ACC-16 ARISTA Access Node Hardware/Environmental Fault SOP

**Node:** IND-ACC-16 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_6450

## Description

The Arista Access Node (IND-ACC-16) at the Indiranagar Exchange site is a critical component of the overall network architecture, providing direct connectivity to the internet and internal network resources. The Access Node operates within the context of the Arista Extensible Operating System (EOS), utilizing a proprietary ASIC chipset to facilitate high-speed packet forwarding and processing. The EOS software stack, built on top of the Arista proprietary ASIC, relies heavily on the line card components to provide the necessary I/O interfaces for connecting to external network links. The power supply redundancy, a critical aspect of the overall system reliability, ensures continued operation in the event of a single power supply failure. However, the loss of both power supplies can lead to a catastrophic failure of the system, resulting in the propagation of a Major alarm (ALARM_6450) due to the loss of critical system components and the inability to maintain network connectivity. This SOP aims to provide a structured approach to identifying the root cause of the hardware/environmental fault and resolving the issue, ensuring minimal downtime and maintaining overall network availability.

## Pre-requisite Checks

- Verify that the Indiranagar Exchange site has been notified and that the necessary remote hands are dispatched to assist with the issue.

- Confirm that all redundant links are operational and that no other network issues are present.

- Ensure that the Arista Access Node (IND-ACC-16) is physically accessible and that the necessary environmental controls are in place to prevent further damage to the equipment.

## Potential Root Causes

| Cause | Probability |
|---|---|
| ASIC memory parity error | High |
| Line card crash | Medium |
| Power supply redundancy loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show system internal memory, Expected Output: Memory information, including any errors or warnings, Description: This command provides detailed information about the system memory, including any errors or warnings that may indicate a hardware fault. The output will help identify any memory-related issues that may be contributing to the failure.
```

```
Command: show hardware linecard, Expected Output: Line card information, including status and error messages, Description: This command provides detailed information about the line card components, including their status and any error messages that may indicate a hardware fault. The output will help identify any line card-related issues that may be contributing to the failure.

Command: show power supply, Expected Output: Power supply information, including status and error messages, Description: This command provides detailed information about the power supply components, including their status and any error messages that may indicate a hardware fault. The output will help identify any power supply-related issues that may be contributing to the failure.
```

## Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace system internal memory [asic_name], Description: This command initiates the replacement of the faulty ASIC. The specific ASIC name should be obtained from the diagnostic output. After replacing the ASIC, ensure that the system is restarted and that all components are functioning properly.

Step 2: Step: Replace the faulty line card, Command: replace hardware linecard [linecard_name], Description: This command initiates the replacement of the faulty line card. The specific line card name should be obtained from the diagnostic output. After replacing the line card, ensure that the system is restarted and that all components are functioning properly.

Step 3: Step: Replace the power supply, Command: replace power supply [power_supply_name], Description: This command initiates the replacement of the power supply. The specific power supply name should be obtained from the diagnostic output. After replacing the power supply, ensure that the system is restarted and that all components are functioning properly.

## Post-Check Verification

```
Command: show system internal memory, Expected Output: No errors or warnings in the memory information, Description: This command verifies that the system memory is functioning properly and that there are no errors or warnings present.

Command: show hardware linecard, Expected Output: All line card components are functioning properly and there are no error messages, Description: This command verifies that the line card components are functioning properly and that there are no error messages present.

Command: show power supply, Expected Output: All power supply components are functioning properly and there are no error messages, Description: This command verifies that the power supply components are functioning properly and that there are no error messages present.
```

## Roll-back Procedures

- Step: Restore the original ASIC, Command: rollback system internal memory [asic_name], Description: This command restores the original ASIC. The specific ASIC name should be obtained from the diagnostic output. After rolling back the ASIC, ensure that the system is restarted and that all components are functioning properly.

- Step: Restore the original line card, Command: rollback hardware linecard [linecard_name], Description: This command restores the original line card. The specific line card name should be obtained from the diagnostic output. After rolling back the line card, ensure that the system is restarted and that all components are functioning properly.

- Step: Restore the original power supply, Command: rollback power supply [power_supply_name], Description: This command restores the original power supply. The specific power supply name should be obtained from the diagnostic output. After rolling back the power supply, ensure that the system is restarted and that all components are functioning properly.